Unit 13, Lesson 2: Compound Interest

Benchmarks

MA.912.FL.3.2 Solve real-world problems involving simple, compound, and continuously compounded interest.

MA.912.FL.3.4 Explain the relationship between simple interest and linear growth. Explain the relationship between compound interest and exponential growth, and the relationship between continuously compounded interest and exponential growth.

MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.

Learning Targets

- ☐ I can explain the relationship between compound interest and exponential growth.
- ☐ I can solve real-world problems involving compound interest.
- ☐ I can compare simple interest growth and compound interest growth.

13.2.1 Warm-Up: Paramedic

1. Kelly Grayson is an emergency medical technician (EMT). In an interview, she was asked, "How do you stay focused and deal with the crisis situation to solve the problem at hand when responding to an emergency call?" Kelly Grayson replied, "You focus on process and fall back on your training."

   a. How does staying focused on the process and falling back on your prior training help you in difficult situations?

   b. What is one thing you have done to help you when learning difficult concepts in mathematics?

13.2.2 Collaboration: Understanding Compound Interest

Compound interest is a method of computing interest. Interest is computed from the (original) principal and the interest earned during a certain time period.

The formula $A = P\left(1+\frac{r}{n}\right)^{nt}$ is used to calculate the final amount owed or saved, where A = final amount, P = principal or original amount, t = number of years, n = number times per year the interest is being compounded, and r = rate of interest per year.

When money is invested, compound interest means that interest is being accrued based on both the principal and previous interest earned.

When money is owed, such as a credit card bill, compound interest means that interest is owed based on both the principal and previous interest accrued.

1. Jasmyne invests $\$10,000$ into a savings account with an interest rate of 6% annual percentage rate (APR) that is compounded quarterly.

   The table shows the amount of Jasmyne's savings for different compounding periods after 1 and 10 years.

Compounding Period	Amount After 1 Year	Amount After 10 Years
quarterly	$10,613.64	$18,140.18
monthly	$10,616.78	$18,193.97
daily	$10,618.31	$18,220.29

   a. What do you notice about the table?

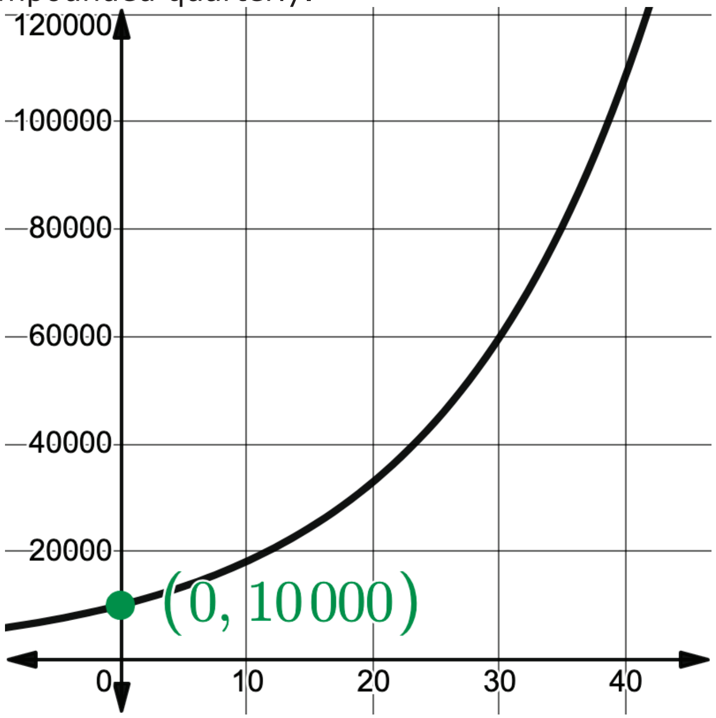
   b. Discuss with your partner why the amount after 1 year is different for each compounding period. Summarize your discussion.

2. Work with your partner to explain why compound interest for investments, such as savings accounts or certificates of deposits (CD), are better than simple interest.

3. Work with your partner to explain why compound interest for loans, such as a car loan, is not better than simple interest.

13.2.3 Guided Instruction: Compound Interest

1. The graph shows an initial investment of $\$10,000$ in a savings account. The interest rate is 6% compounded quarterly.



a. What do you notice about the graph?

b. What type of function best represents the graph?

○ linear
○ quadratic
○ exponential

c. Discuss with your partner how the formula for compound interest, $A = P\left(1+\frac{r}{n}\right)^{nt}$, supports your function choice. Summarize your discussion.

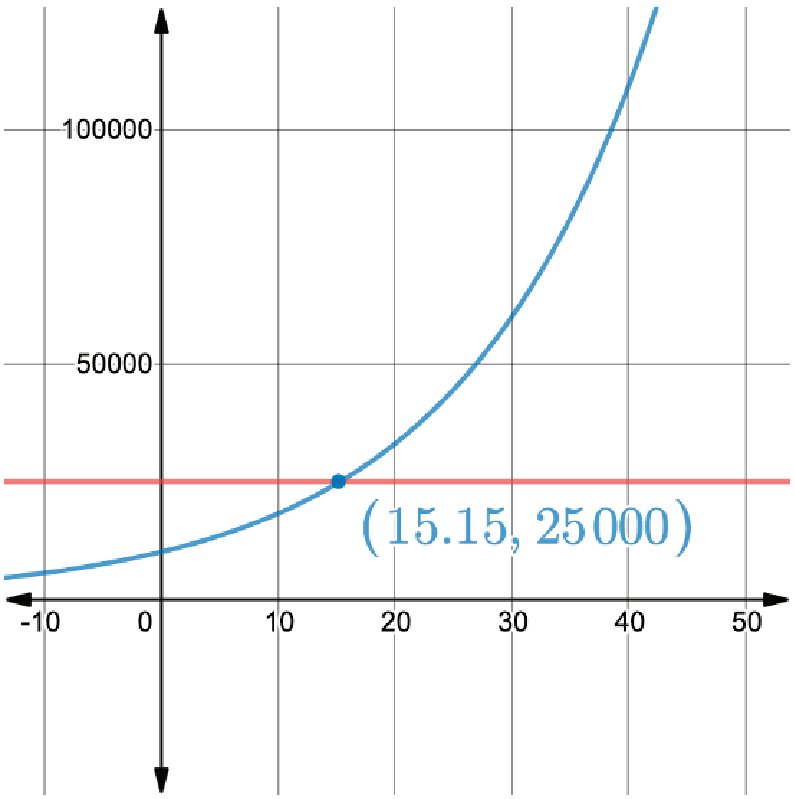
d. Complete the table.

Key Feature	Description
y-intercept	
End Behavior	As $x \to \infty$, $y \to$
Increasing/ Decreasing	As x increases, $y \to$

e. Use the information about the graph to write an equation that models the relationship.

f. What is the balance of the savings account after 7 years?

g. Use the graph shown to determine when the savings account will reach $\$25{,}000$.



2. Amir deposited $\$32,500$ in a savings account with a 4.25% APR compounded monthly.

   a. Write the equation that models the amount of money Amir has in the account.

   b. Use the equation to complete the table. Show your work.

Number of Years	Total Amount Saved
5	
35	

   c. Discuss with your partner the relationship between compound interest and exponential growth. Summarize your discussion.

13.2.4 Collaboration: Comparing Interest

1. Work with your partner to complete the table to compare two investments.

	Simple Interest	Compound Interest
Investment Amount and APR	$\$50,000$ is invested at an interest rate of 2.75% APR	$\$50,000$ is invested at an interest rate of 2.75% APR compounded monthly
Write the equation.	$A = P(1 + rt)$	$A = P\left(1 + \frac{r}{n}\right)^{nt}$
Determine the amount after 10 years.		
Determine the amount after 20 years.		

13.2.5 Wrap-Up: Ticket Out the Door

1. Complete the table to compare two investments.

	Simple Interest	Compound Interest
Investment Amount and APR	$\$2{,}325$ is invested at an interest rate of 4.5% APR	$\$2{,}325$ is invested at an interest rate of 4.5% APR compounded monthly
Write the equation.	$A = P(1 + rt)$	$A = P\left(1 + \frac{r}{n}\right)^{nt}$
Determine the amount after 8 years.		

Unit 13, Lesson 3: Comparing Key Features of Functions

Benchmarks

MA.912.F.1.6 Compare key features of linear and nonlinear functions each represented algebraically, graphically, in tables, or written descriptions.

MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.

Learning Targets

- ☐ I can compare key features of linear and nonlinear functions that are represented algebraically, graphically, in tables, or in written descriptions.
- ☐ I can verify when a quantity increasing exponentially exceeds a quantity increasing linearly or quadratically.